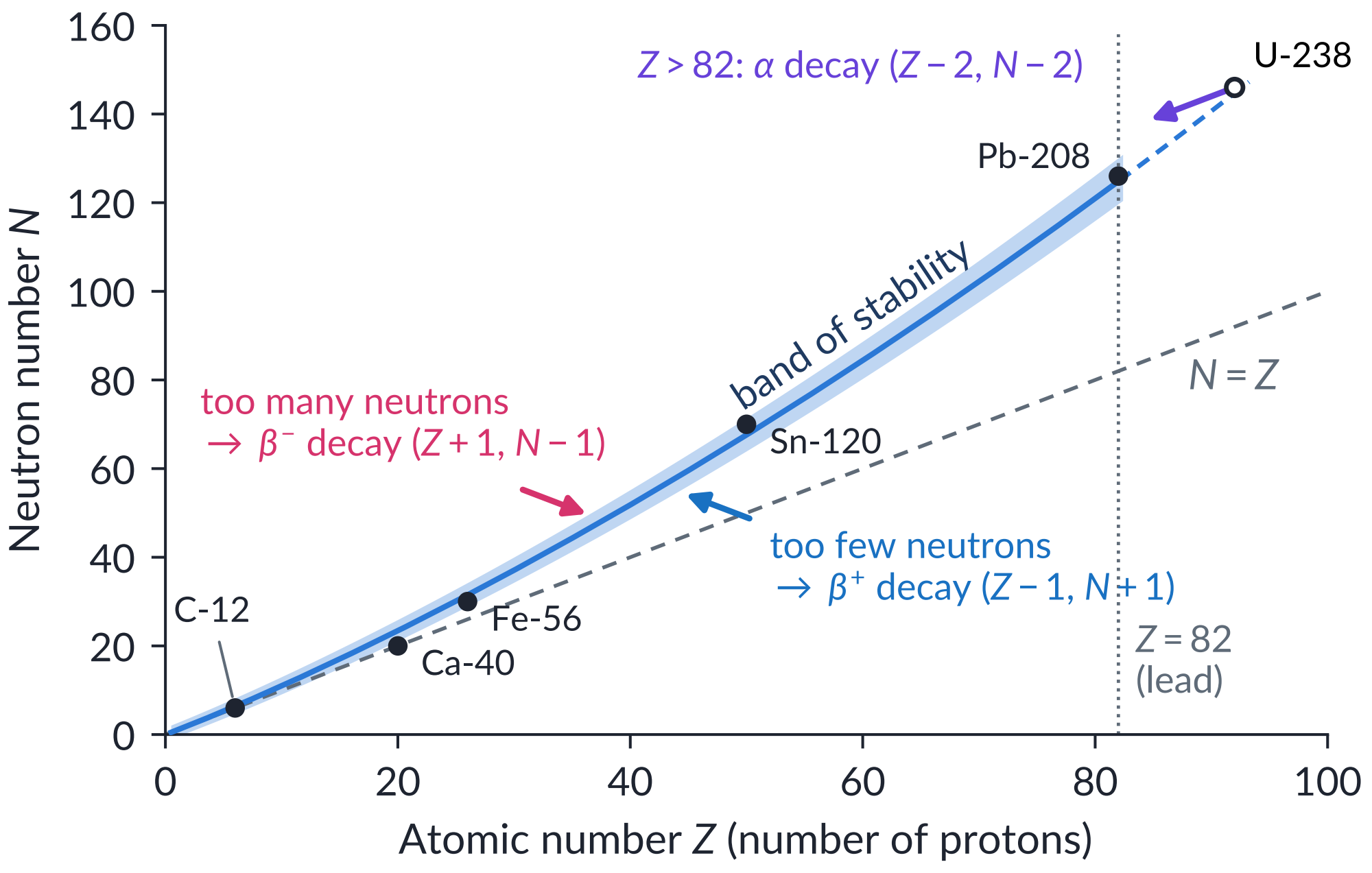
Neutron number N
Atomic number Z (number of protons)
0
20
40
60
80
100
120
140
160
Z > 82: α decay (Z − 2, N − 2)
U-238
Pb-208
band of stability
N = Z
too many neutrons
→ β⁻ decay (Z + 1, N − 1)
Sn-120
too few neutrons
→ β⁺ decay (Z − 1, N + 1)
C-12
Fe-56
Ca-40
Z = 82
(lead)